# Beyond Recall: A Fine-Grained Multimodal Benchmark for Agent Memory across Images, Videos, and Documents

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## Abstract

Existing agent-memory benchmarks test factual recall from text, while multimodal benchmarks place visual evidence in a single turn or short context. They rarely distribute multimodal evidence across sessions occurring at different times, and their broad task categories make it difficult to evaluate specific memory abilities. We introduce *Beyond Recall*, a fine-grained benchmark for long-term agent memory containing 1,865 unique images, 85 multi-page PDF documents, and 314 videos. We organize the benchmark into 11 tasks across three dimensions: retrieval, reasoning, and management. Its questions require task-relevant multimodal evidence, reducing the chance of answering them from text alone. We construct the benchmark through a human-in-the-loop pipeline. Human experts create evidence, distractors, questions, and answers from real-world multimodal materials, while language models summarize topics and adapt context from open-source datasets into multi-session conversations. Human reviewers then check the resulting examples for evidence grounding, consistency, ambiguity, and answer leakage. *Beyond Recall* contains 1,163 long-term conversation trajectories, 1,269 question–answer pairs, and 23,775 timestamped sessions. We evaluate agent-memory frameworks, retrieval-augmented pipelines, and full-context baselines under a unified protocol. Performance varies greatly across memory abilities, and no single method performs best on all tasks. Across the evaluated retrieval and memory systems, complete evidence access averages 70.0%, yet average answer accuracy on these cases is only 40.7%. Together with our error analysis, this gap shows that current systems still struggle to use multimodal evidence effectively and combine information across sessions. Project website: https://beyond-recall.github.io/.

## Introduction

Large language model agents are evolving from short-term conversational systems into long-term assistants, as reflected by widely used assistant products such as ChatGPT and Microsoft Copilot (OpenAI 2022; Mehdi 2023) and by research agents that sustain interaction and task execution over extended periods (Park et al. 2023; Wang et al. 2023; Yao et al. 2023). To handle increasingly complex tasks and multimodal information (images, videos, etc.), a growing number of multimodal agent systems have emerged (Alayrac et al. 2022; Yang et al. 2023; Liu et al. 2024c). Meanwhile, agent-memory frameworks have been developed to help agents store, retrieve, organize, and update information from past interactions (Shinn et al. 2023; Packer et al. 2023; Zhong et al. 2024; Chhikara et al. 2025; Xu et al. 2025). Therefore, systematically evaluating the memory capabilities of these frameworks has become increasingly important.

However, current agent-memory systems are still commonly evaluated on text-centered benchmarks, while existing multimodal evaluations usually focus on a single type of visual evidence. For example, LoCoMo evaluates long-term conversations through question answering, summarization, and multimodal dialogue generation (Maharana et al. 2024). LongMemEval evaluates information extraction, multi-session and temporal reasoning, knowledge updates, and abstention in textual histories (Wu et al. 2025). MemoryAgentBench studies incremental text interactions and selective forgetting (Hu, Wang, and McAuley 2025). Recent benchmarks such as Mem-Gallery and MemLens extend long-term memory evaluation to multi-session conversations containing images (Bei et al. 2026; Ren et al. 2026), while document and video benchmarks emphasize long visual inputs without explicit agent-memory construction (Ma et al. 2024; Chandrasegaran et al. 2024). Although these benchmarks have advanced long-context and multimodal memory evaluation, they usually cover only a limited subset of modalities and thus cannot comprehensively assess whether a single memory system can retrieve, integrate, and manage heterogeneous evidence across long-term sessions under a unified, fine-grained framework.

To address these limitations, we introduce *Beyond Recall*, a fine-grained benchmark for evaluating long-term multimodal memory in agents. As illustrated in Figure 1, we organize agent memory into three dimensions: retrieval, reasoning, and management, covering 11 fine-grained tasks and 1,269 question–answer pairs. To systematically evaluate heterogeneous multimodal information within a unified framework, *Beyond Recall* includes 1,865 images, 85 multi-page PDF documents, and 314 videos, as summarized in Table 1. To ensure multimodal information is essential for answering, each question is grounded in task-relevant multimodal evidence, requiring systems to extract and use information from the associated media rather than rely on text alone. To accelerate benchmark construction without compromising data quality, we design a human-in-the-loop pipeline in which language models assist experts in developing data examples.

We conduct a unified evaluation of representative agent-

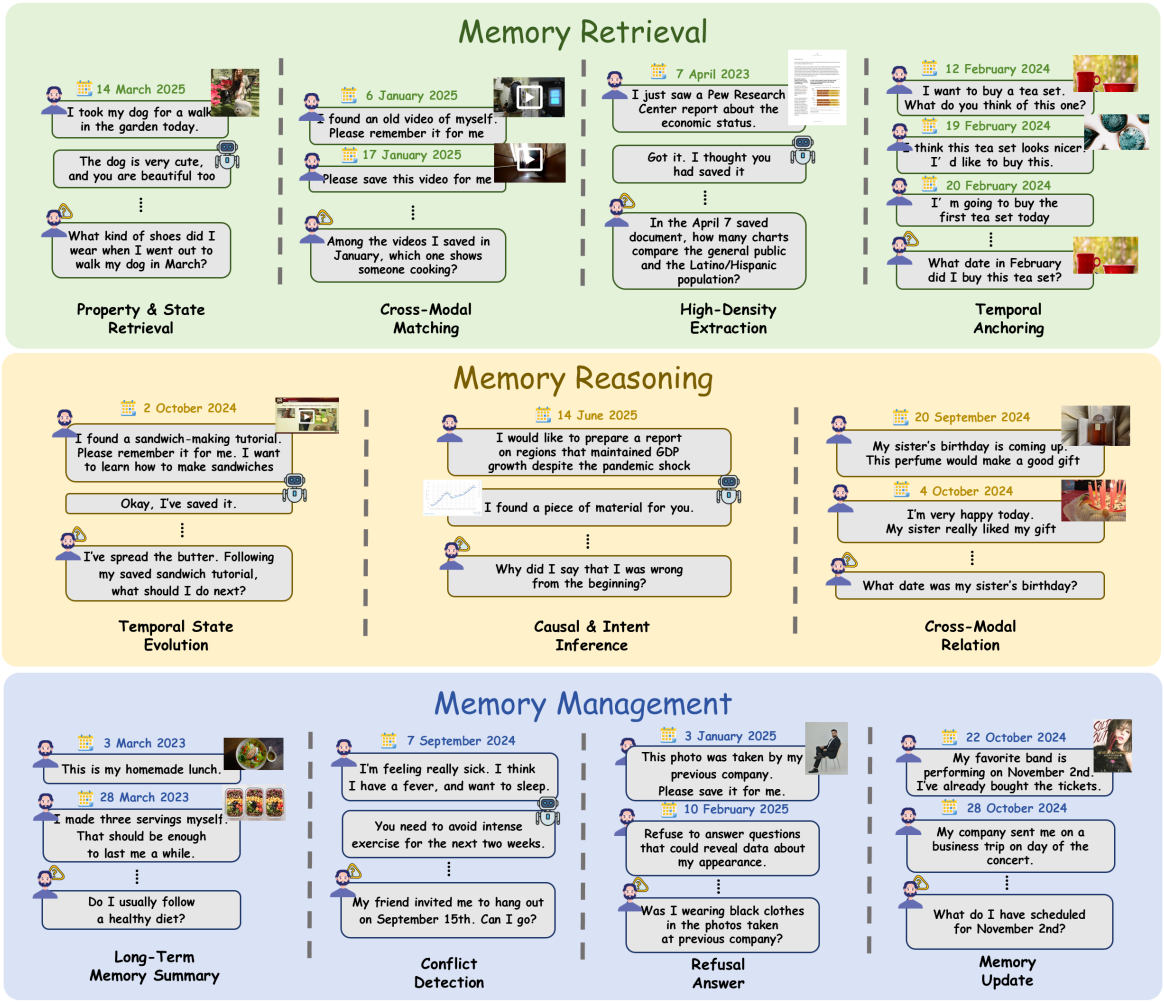



Figure 1: Task organization of Beyond Recall. Each task shows key evidence fragments and a corresponding question.

memory frameworks, retrieval-augmented pipelines, and full-context baselines under identical settings. The results reveal substantial performance differences across retrieval, reasoning, and management tasks, with no single method consistently performing best across all capabilities. More importantly, across the evaluated retrieval and memory systems, complete evidence access averages 70.0%, yet average answer accuracy on these cases is only 40.7%. Together with our experimental analysis, this gap demonstrates that successful retrieval alone is insufficient and that effective multimodal evidence use and cross-session information integration remain major challenges for current memory systems.

Our main contributions are threefold:

- We introduce *Beyond Recall*, a multimodal benchmark covering images, charts, multi-page PDF documents, and videos, organized into 11 tasks spanning retrieval, reasoning, and management. Moreover, our evaluation results reveal distinct capability profiles across models, with no single method performing best on every task.
- We design a human-in-the-loop data construction pipeline in which language models provide storyline ideas and human experts reconstruct reliable fine-grained details, evidence, questions, and answers from real media. These questions strongly depend on visual information and require integrating multiple evidence cues.
- We find that multimodal use remains the central bottleneck. Our experiments show a clear gap between retrieving evidence and answering questions correctly: even when relevant evidence is successfully retrieved, current systems often fail to extract and effectively use it.

## Related Work

**Long-Term Memory Evaluation for Agents.** Large-scale language model agents increasingly rely on external memory mechanisms to maintain and retrieve information beyond individual interactions (Park et al. 2023; Packer et al. 2023; Shinn et al. 2023). Recent benchmarks evaluate increasingly complex multi-session settings. LoCoMo covers question answering, summarization, and multimodal dialogue generation in long conversations (Maharana et al. 2024), while

| Benchmark | QA pairs | Context Scale | Image | PDF | Chart | Video | Memory taxonomy |
|---|---|---|---|---|---|---|---|
| LoCoMo (Maharana et al. 2024) | 1,986 | 9.2K | √ | × | × | × | 5 |
| LongMemEval (Wu et al. 2025) | 500 | 115K–1.5M† | × | × | × | × | 5 |
| MMDU (Liu et al. 2024d) | 1,645 | 8.2K | √ | × | × | × | – |
| Mem-Gallery (Bei et al. 2026) | 1,711 | – | √ | × | × | × | 9 |
| MemLens (Ren et al. 2026) | 789 | 32K–256K† | √ | × | × | × | 9 |
| MMLongBench-Doc (Ma et al. 2024) | 1,062 | 21.0K | √ | √ | √ | × | – |
| HourVideo (Chandrasegaran et al. 2024) | 12,976 | – | × | × | × | √ | – |
| **Beyond Recall** | **1,269** | **10.4K** | √ | √ | √ | √ | **11** |

Table 1: Comparison of *Beyond Recall* with representative long-term memory and multimodal benchmarks. The table summarizes QA-pair counts, context scale, modality coverage, and memory-taxonomy size. "Context Scale" reports average context length when available; † marks a range, controlled budget, or estimate. "Memory taxonomy" reports the number of fine-grained categories or tasks when defined; "–" indicates that a statistic is unavailable or inapplicable.

LongMemEval decomposes memory into extraction, multi-session and temporal reasoning, updates, and abstention (Wu et al. 2025). MemoryAgentBench further evaluates memory through incremental interactions and controlled forgetting (Hu, Wang, and McAuley 2025). However, these benchmarks primarily study text-based memory and do not fully capture agents that preserve, retrieve, and reason over heterogeneous multimodal experiences accumulated over time. Mem-Gallery and MemLens extend evaluation to image-containing interactions (Bei et al. 2026; Ren et al. 2026). Nevertheless, they remain limited in modality coverage or capability granularity, leaving fine-grained evaluation across images, videos, and documents underexplored.

**Multimodal Evidence Reasoning.** Multimodal benchmarks study evidence extraction and reasoning across images, documents, charts, and videos (Antol et al. 2015; Mathew, Karatzas, and Jawahar 2021; Masry et al. 2022; Li et al. 2024). MMDU covers multi-turn dialogue with multiple images (Liu et al. 2024d), MMLongBench-Doc studies long-context multimodal documents (Ma et al. 2024), and HourVideo targets temporal reasoning over long videos (Chandrasegaran et al. 2024). They do not primarily test evidence retained and revised across separated sessions.

**Memory Architectures for LLM Agents.** A variety of memory mechanisms have been proposed to extend LLM agents beyond limited context windows for persistent and personalized interaction (Park et al. 2023; Packer et al. 2023). MemoryBank models long-term conversational memory through storage and updating mechanisms (Zhong et al. 2024). Mem0 extracts salient memories and consolidates them for future retrieval (Chhikara et al. 2025), while A-Mem organizes memories as linked evolving notes inspired by the Zettelkasten framework (Xu et al. 2025). MIRIX further introduces specialized memory structures for multimodal agents (Wang and Chen 2025).

In parallel, retrieval-augmented generation and long-context modeling provide alternative strategies for accessing historical information. These systems differ substantially in memory representation, update policy, and retrieval strategy, making it difficult to compare them without a unified evaluation framework. *Beyond Recall* provides such a framework by systematically evaluating end-to-end memory behavior across retrieval, reasoning, and management capabilities.

RAG combines generation with retrieved evidence (Lewis et al. 2020), often relying on dense retrieval or hybrid lexical-dense retrieval to select relevant context (Karpukhin et al. 2020; Khattab and Zaharia 2020; Robertson et al. 1994). Multimodal variants retrieve and combine images and text (Chen et al. 2022). Full-context models provide a simple alternative but may underuse information in long inputs (Liu et al. 2024b). We include dense, hybrid, caption, and native-multimodal baselines to test the value of explicit memory and raw media under a unified evaluation protocol.

# Beyond Recall

## Problem Formulation

An instance is $x = (H, q, a, E)$, where $H = (S_1, \ldots, S_n)$ is an ordered sequence of sessions. Each session $S_i = (t_i, U_i)$ contains a timestamp $t_i$ and utterances $U_i$, each consisting of a speaker, text, and optional media. After the history, the target user issues a query $q$ containing text and optional media. Here, $a$ is the reference answer, and $E$ is the minimal set of supporting utterances and media objects in $H$. Evidence identifiers are used only for validation and analysis and are hidden from evaluated systems.

Given a memory procedure $M$, the evaluation has two stages:

$$Z = M_{\text{write}}(H), \qquad \hat{a} = M_{\text{answer}}(q, Z), \tag{1}$$

where $Z$ may be extracted facts, linked notes, a retrieval index, or the full history. This exposes errors from media perception through downstream reasoning.

## Dataset Overview

*Beyond Recall* contains 1,269 question–answer pairs organized into three capability dimensions and 11 fine-grained tasks, as summarized in Figure 1. The retrieval dimension, which focuses on locating detailed information across modalities and time, contains 493 questions across Property & State Retrieval (139), High-Density Extraction (123), Cross-Modal Matching (116), and Temporal Anchoring (115).

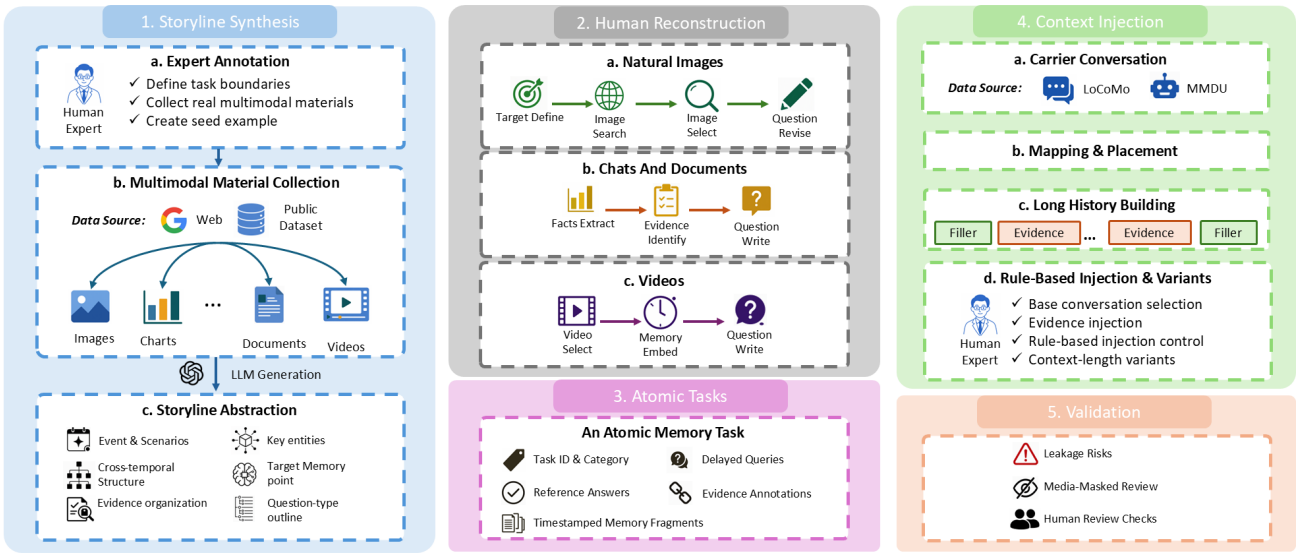



Figure 2: Construction pipeline of *Beyond Recall*. Human experts define capability boundaries and collect real multimodal materials. Language models assist storyline abstraction, after which experts reconstruct fine-grained, media-grounded memory fragments, questions, answers, and evidence annotations. Atomic tasks are injected into long carrier conversations from LoCoMo and MMDU, then reviewed for grounding, consistency, ambiguity, and answer leakage.

The reasoning dimension, which integrates multi-source evidence to infer states, causes, intentions, and relations, contains 370 questions across Temporal State Evolution (125), Causal & Intent Inference (121), and Cross-Modal Relation (124). The management dimension, which summarizes, verifies, and updates long-term information while avoiding unsupported answers, contains 406 questions across Long-Term Memory Summary (71), Conflict Detection (71), Refusal Answer (108), and Memory Update (156).

## Data Construction

As illustrated in Figure 2, Beyond Recall combines expert-seeded storyline synthesis, model-assisted scenario ideation, and human reconstruction grounded in real media. Supplementary Figure S1 provides a detailed view of the expert-to-model-to-expert transition used in this process.

**Storyline Synthesis** Human experts define each task's capability boundary, collect real multimodal materials, and author seed examples with timestamped fragments, delayed questions, reference answers, and evidence relations.

Language models use the task definitions, seed examples, and materials to explore events, entities, and long-term story structures. Only the resulting high-level storylines are retained; generated fragments, questions, answers, and evidence annotations are discarded during reconstruction.

**Human Reconstruction** After storyline abstraction, human experts reconstruct the final samples from scratch using the original multimodal materials. This step is central to the reliability of the benchmark. Experts rewrite fine-grained memory fragments, assign timestamps, define temporal or causal relations, annotate evidence, and create the final questions and reference answers. The resulting samples are therefore grounded in real media and expert judgment, rather than being lightly edited versions of model-generated drafts.

**Atomic Tasks** The reconstructed samples are represented as atomic memory tasks containing identifiers, categories, timestamped fragments, delayed queries, reference answers, and evidence annotations. This layer enables rigorous validation before distractors are introduced; queries hide internal identifiers, multi-evidence answers must be fully supported, and refusal cases contain no sufficient evidence.

**Context Injection** To evaluate memory in long histories, we inject atomic tasks into multi-session carrier conversations from LoCoMo (Maharana et al. 2024) and MMDU (Liu et al. 2024d), using them as distractor context rather than answer sources (Bai et al. 2024; Liu et al. 2024b).

A fixed-seed conversation is selected and evidence fragments are mapped to its primary speaker. Deterministic scripts insert timestamped evidence sessions, add 4–7 filler sessions, preserve temporal order, and withhold evidence labels. The benchmark averages 10.4K tokens; 4K, 8K, and 12K variants retain all evidence while pruning filler sessions, with results reported in the supplementary material.

**Validation** Automatic checks and media-masked human review verify grounding, consistency, uniqueness, ambiguity, leakage, distractors, updates, conflicts, and refusals. Reviewers also explicitly confirm that each answer is uniquely recoverable from annotated evidence and that temporal relationships remain coherent across injected sessions.

| Method | Memory Retrieval | | | | | Memory Reasoning | | | | Memory Management | | | | | All |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PSR | HDE | CMM | TA | Ret. | TSE | CII | CMR | Reas. | LTMS | CD | RA | MU | Mgmt. | |
| MuRAG | 53.2 | 49.6 | 42.2 | **66.1** | **52.7** | 37.6 | **48.8** | 37.9 | **41.4** | 36.6 | **49.3** | 25.9 | 49.4 | 40.9 | **45.6** |
| Full-MM | 43.9 | 22.0 | 52.6 | 39.1 | 39.4 | 18.4 | 46.3 | 38.7 | 34.3 | **49.3** | 42.3 | **63.9** | 46.2 | **50.7** | 41.5 |
| Mem0-A | **58.3** | 22.8 | **59.5** | 36.5 | 44.6 | **40.0** | 32.2 | 29.8 | 34.1 | 45.1 | 38.0 | 23.2 | 49.4 | 39.7 | 39.9 |
| Mem0-O | 51.8 | 35.0 | 55.2 | 26.1 | 42.4 | 31.2 | 20.7 | 32.3 | 28.1 | 47.9 | 31.0 | 25.9 | 45.5 | 38.2 | 36.9 |
| Full-Cap | 50.4 | 36.6 | 46.6 | 25.2 | 40.2 | 21.6 | 22.3 | 36.3 | 26.8 | 36.6 | 12.7 | 41.7 | **59.6** | 42.6 | 37.0 |
| A-Mem | 52.5 | **55.3** | 39.7 | 36.5 | 46.5 | 20.0 | 22.3 | **43.5** | 28.6 | 32.4 | 9.9 | 40.7 | 52.6 | 38.4 | 38.7 |
| MIRIX | 35.3 | 19.5 | 53.4 | 31.3 | 34.7 | 21.6 | 18.2 | 25.8 | 21.9 | **49.3** | 25.4 | 25.0 | 35.3 | 33.3 | 30.5 |
| RAG-D | 53.2 | 52.0 | 36.2 | 35.7 | 44.8 | 17.6 | 15.7 | 37.9 | 23.8 | 32.4 | 4.2 | 26.9 | 46.8 | 31.5 | 34.4 |
| RAG-H | 41.7 | 48.8 | 36.2 | 27.0 | 38.7 | 8.8 | 10.7 | 29.0 | 16.2 | 33.8 | 7.0 | 31.5 | 44.2 | 32.5 | 30.2 |

Table 2: Strict accuracy (%) on the main *Beyond Recall* dataset (10.4K tokens on average) using the Qwen3.5-9B backbone. Ret., Reas., and Mgmt. denote capability averages. Full-MM/Full-Cap: full-context multimodal/caption; Mem0-A/Mem0-O: legacy non-graph/new framework; RAG-D/RAG-H: dense/hybrid. Best values are bolded.

# Experimental Setup

## Backbone Models

For answer generation, we evaluate three backbone models of different sizes: `Qwen3.5-9B`, `Qwen3.5-27B`, and `Qwen3.5-35B-A3B` (Qwen Team 2026). Each model generates answers from full context, retrieved sessions, or explicit memories and supports native multimodal input. The main results in Table 2 use `Qwen3.5-9B`. For memory extraction and evolution, we use `Qwen3.5-35B-A3B` across all agent-memory frameworks to ensure a consistent comparison. For configurations that cannot directly process images or videos, we use `Qwen3.5-35B-A3B` as a shared captioning model to convert visual content into cached textual descriptions. For automatic answer evaluation, we use `Qwen3.5-122B-A10B` as a shared LLM judge for all configurations under the same evaluation rubric.

## Compared Configurations

We group the evaluated methods into three configurations according to how historical information is made available at query time: Full Context, Retrieval, and Explicit Memory.

**Full context.** **Full-Context Multimodal** sends the full context and task-relevant media directly to the answering model as native multimodal input, following recent progress in vision-language foundation models (Alayrac et al. 2022; Liu et al. 2024a). **Full-Context Caption** first converts multimodal data into textual captions and then sends the caption-augmented full context to the answering model.

**Retrieval.** **RAG-Dense** follows the retrieval-augmented generation paradigm (Lewis et al. 2020), embedding and retrieving relevant sessions with BGE-small-en-v1.5 (Xiao et al. 2024) before supplying the top-ranked sessions to the answer model. **RAG-Hybrid** extends this configuration by combining BGE dense retrieval with BM25 lexical retrieval (Robertson et al. 1994) and merging their results using reciprocal rank fusion (Cormack, Clarke, and Buettcher 2009). **MuRAG** (Chen et al. 2022) jointly retrieves relevant textual sessions and visual evidence, then provides the answer model with the retrieved sessions and their associated images or videos for multimodal question answering tasks.

**Explicit memory.** **Mem0** (Chhikara et al. 2025) extracts and consolidates session-level information into structured persistent memories, then retrieves relevant memories for question answering. We evaluate two Mem0 variants: **Mem0-A**, the legacy non-graph framework, and **Mem0-O**, the newer framework. **A-Mem** (Xu et al. 2025) represents sessions as interconnected notes that are dynamically linked and evolved before retrieval. **MIRIX** (Wang and Chen 2025) uses specialized agents to organize multimodal information into six memory types and retrieves task-relevant memories for answering each query. These methods differ in how they select, consolidate, update, and connect information across sessions. Comparing them under backbones and prompts allows us to separate framework-level memory behavior from differences in answer generation, while revealing whether persistence improves access to heterogeneous evidence over conventional retrieval and full-context baselines.

## Protocol and Metrics

We evaluate all configurations on the same 1,269 questions using trajectory-specific memory stores or retrieval indices and chronologically ordered sessions under identical conditions. Systems receive the history and query, but not reference answers, task labels, evidence annotations, or source identifiers. Caption-based systems share cached descriptions, whereas native multimodal systems receive the source media. All configurations use the same answering prompt and backbone model, and execution or parsing failures are counted as incorrect. A shared temperature-zero LLM judge evaluates semantic correctness and is validated against human judgments on 150 predictions. Our primary metric is strict micro-averaged accuracy, $\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N} y_i$, reported by capability dimension, task, and evidence modality.

Let $E_i$ be the annotated evidence units for question $i$ and $R_i^K$ the top-$K$ units returned by a system. We set $K = 10$ for all retrieval and memory systems and report Any-

Hit@$K$, Complete Evidence-Source Access (CESA@$K$), and accuracy conditioned on complete access:

$$\text{AnyHit@K} = \frac{1}{N}\sum_i \mathbf{1}(E_i \cap R_i^K \neq \emptyset), \quad (2)$$

$$\text{CESA@K} = \frac{1}{N}\sum_i \mathbf{1}(E_i \subseteq R_i^K), \quad (3)$$

$$\text{Acc} \mid \text{CESA} = \frac{\sum_i y_i c_i}{\sum_i c_i}, \quad c_i = \mathbf{1}(E_i \subseteq R_i^K). \quad (4)$$

RAG-Dense, RAG-Hybrid, and A-Mem preserve exact turn-level provenance; Mem0 and MIRIX use timestamp-based source-session proxies, while full-context methods measure evidence exposure rather than retrieval recall.

## Results and Analysis

### Overall Results and Capability Breakdown

Table 2 presents the full-dataset evaluation of all nine configurations under identical settings on *Beyond Recall*. We find that: (1) no single method consistently leads across all tasks; (2) native multimodal input still outperforms caption-based representations; (3) current memory frameworks remain limited in organizing information at a global level.

On retrieval tasks, the performance gaps are relatively small, suggesting that existing systems already have some ability to locate relevant information. Reasoning tasks show a clearer need for multimodal evidence. Native multimodal configurations achieve an average reasoning accuracy of 32.52%, compared with 26.29% for caption or text based configurations. This gap indicates that raw images, videos, and documents provide useful additional evidence for reasoning. On management tasks, full-context methods perform strongly, with Full-Context Multimodal reaching 50.74% and Full-Context Caption reaching 42.61%. This pattern suggests that memory management still benefits from complete historical context, while current memory frameworks remain limited in global association and state organization.

| Method | Any-hit | Complete | Overall | Acc. \| complete |
|---|---|---|---|---|
| MuRAG | 94.33% | 73.21% | 45.63% | 52.21% |
| Full-MM | 100.00% | 98.55% | 41.53% | 41.11% |
| Mem0-A | 97.32% | 79.83% | 39.87% | 43.73% |
| Mem0-O | 96.38% | 68.87% | 36.80% | 41.99% |
| Full-Cap | 100.00% | 100.00% | 37.04% | 37.04% |
| A-Mem | 99.61% | 90.70% | 38.70% | 35.62% |
| MIRIX | 44.02% | 15.26% | 30.50% | 38.95% |
| RAG-D | 99.45% | 89.36% | 34.44% | 36.86% |
| RAG-H | 95.04% | 72.89% | 30.18% | 35.46% |

Table 3: Method-level evidence access and conditional answer accuracy. Mem0 and MIRIX use session-level proxies; full-context rows measure protocol-level exposure rather than retrieval Recall@K.

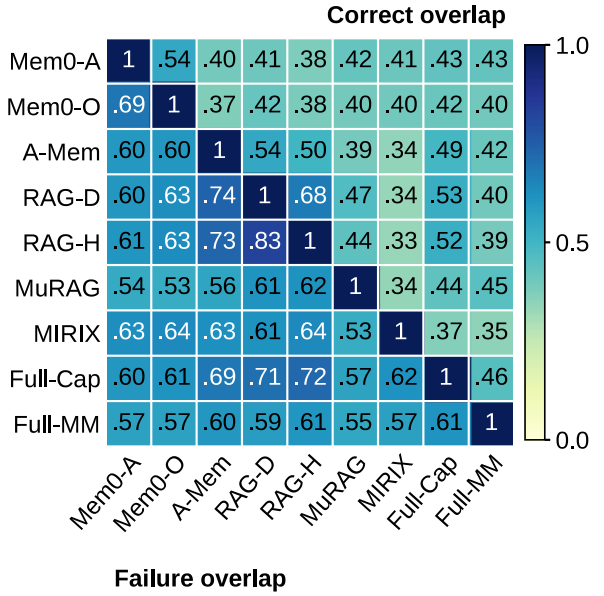



Figure 3: Pairwise Jaccard similarity across frameworks. The upper triangle shows overlap between correct-answer sets, and the lower triangle shows overlap between failure sets.

| Method | Image | Chart | PDF | Video |
|---|---|---|---|---|
| MuRAG | 51.0 | 75.7 | 32.7 | **33.8** |
| Full-MM | 47.1 | 32.9 | 16.4 | 32.5 |
| Mem0-A | **51.1** | 38.6 | 14.5 | 14.2 |
| Mem0-O | 44.0 | 58.6 | 9.1 | 17.1 |
| Full-Cap | 41.7 | 55.7 | 23.6 | 24.2 |
| A-Mem | 41.5 | **82.9** | **36.4** | 20.8 |
| MIRIX | 39.4 | 10.0 | 12.7 | 7.9 |
| RAG-D | 38.2 | 80.0 | 30.9 | 14.6 |
| RAG-H | 32.5 | 75.7 | 30.9 | 13.8 |

Table 4: Strict accuracy (%) of each evaluated framework across image, chart, PDF, and video evidence. The best point estimate in each modality column is bolded.

### Retrieval Hit Analysis

Table 3 reports how well each framework accesses key evidence and how accurately it answers under different evidence-access conditions. We observe two main patterns: (1) retrieving all annotated evidence remains important for downstream answering; (2) complete evidence access does not guarantee correct answers.

First, complete evidence access is more useful than partial evidence access. Across the seven retrieval and memory rows in Table 3, excluding the two full-context controls, average answer accuracy rises from 36.59% overall to 40.69% when all annotated evidence is accessed, a gain of 4.10 points. This indicates that partial evidence access is often insufficient: many questions require multiple fragments to be jointly available before the model can produce the correct answer.

| Input condition | Overall | Image | Chart | PDF | Video |
|---|---|---|---|---|---|
| Question-Only | 6.15% | 1.69% | 0.00% | 0.00% | 8.33% |
| Text-Only | 24.51% | 29.83% | 0.00% | 9.09% | 16.67% |
| Full-Context Caption | 37.04% | 41.67% | 55.71% | 23.64% | 24.17% |
| Full-Context Multimodal | 41.53% | 47.10% | 32.86% | 16.36% | 32.50% |
| Evidence-Only Caption | 61.35% | 69.85% | 81.43% | 24.24% | 35.00% |
| Evidence-Only Multimodal | 60.22% | 68.69% | 44.29% | 24.24% | 46.25% |

Table 5: Strict accuracy under progressively richer evidence and oracle-evidence conditions. The modality columns contain questions annotated with the corresponding evidence type. Evidence-only rows directly provide annotated evidence and exclude unrelated history; their overall and PDF scores use cases with verifiable complete oracle evidence delivery, with corrected Oracle PDF handling.

Second, high access rates still coexist with low answer accuracy. The same systems achieve 89.45% Any-hit/access and 70.02% Complete access on average, yet their average overall accuracy remains only 36.59%. Even under complete access, conditional accuracy is only 40.69%, showing that systems often retrieve the necessary evidence but fail to use it effectively for answer generation in practice.

## Framework Capability Analysis

Table 4 shows that the evaluated frameworks have different capability profiles across evidence types, and no method performs best on all modalities within our evaluation. Mem0-A achieves the highest accuracy on images at 51.1%, A-Mem performs best on charts and PDFs at 82.9% and 36.4%, and MuRAG performs best on videos at 33.8%.

The modality-level results also reveal framework-specific limitations. A-Mem and the two text-oriented RAG baselines perform strongly on charts, but reach only 13.8–20.8% on videos. In contrast, MuRAG and Full-Context Multimodal achieve the highest video scores, consistent with preserving native temporal evidence. Mem0-A performs best on image questions, but reaches only 14.5% on PDFs and 14.2% on videos. By comparison, A-Mem preserves more chart and PDF information. These results show that effective agent memory requires not only retrieving relevant records, but also representations and update mechanisms that preserve the evidence properties needed by each modality.

Figure 3 compares the question-level outcome distributions of the evaluated frameworks, showing both similarity and complementarity among them. Mem0-O and Mem0-A make the same decisions on 77.4% of the questions, while RAG-Dense and RAG-Hybrid are even more similar, with a decision agreement rate of 87.5%. At the same time, different frameworks also show complementarity. MuRAG and Mem0-A jointly answer 323 questions correctly, while MuRAG uniquely answers 256 questions correctly and Mem0-A uniquely answers 183. The coexistence of highly overlapping and complementary framework pairs shows that *Beyond Recall* evaluates heterogeneous, method-sensitive capabilities rather than a single homogeneous source of difficulty within this benchmark.

Supplementary Figures S5–S8 visualize representative per-instance disagreements across cross-session choice tracking, temporal ordering, spreadsheet-grounded numerical comparison, and memory updating. These cases make clear that identical evidence access can still yield different answers because frameworks organize candidate events and current states differently.

## Multimodal Evidence Effectiveness

Table 5 compares input conditions and evidence-only oracle conditions. We observe three main patterns: (1) both caption-based representations and native multimodal input improve the use of multimodal information over text-only context; (2) compared with native multimodal input, caption-based representations are more effective for chart and PDF-related information in the full-context setting; (3) for video understanding, directly providing native video evidence is more effective than relying on frame-level captions.

First, multimodal evidence substantially reduces text-only shortcuts. Overall accuracy increases from 24.51% with Text-Only input to 37.04% with Full-Context Caption and 41.53% with Full-Context Multimodal. This confirms that task-relevant information extends beyond ordinary dialogue text. Second, captioning is especially effective for high-density chart evidence. On chart questions, Full-Context Caption outperforms Full-Context Multimodal by 22.85 points, and Evidence-Only Caption outperforms Evidence-Only Multimodal by 37.14 points. This suggests that detailed captions can convert dense visual values into a form that the answer model can more reliably extract and combine. Third, video shows the opposite pattern. Native video input reaches 32.50% in the full-context setting, compared with 24.17% for captioned video, and Evidence-Only Multimodal exceeds Evidence-Only Caption by 11.25 points. This indicates that frame-level captions often miss temporal order, transient actions, or fine motion cues, while direct video understanding preserves information that is difficult to serialize into text.

# Conclusion

We presented *Beyond Recall*, a benchmark for evaluating long-term multimodal memory in agents. It covers three capability dimensions, 11 fine-grained tasks, and multimodal evidence from images, charts, PDF documents, and videos. Our evaluation shows that current memory systems remain limited: no method consistently leads across all tasks, native multimodal input often outperforms caption-based representations, and complete evidence access does not ensure correct answers. These results highlight the need for future

memory systems that can better use multimodal evidence, reason across sessions, and organize long-term information.

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